

even using both simultaneously. It's like having two Wi-Fi cards working in tandem. Result: more stable performance, lower latency (great for gaming or video calls), and better use of available spectrum (especially in crowded environments, the adapter can switch to the least congested link or use both).

4. Other improvements include enhanced OFDMA and MU-MIMO (16 streams) for more capacity in dense networks, and tricks like preamble puncturing (allows using parts of the spectrum that aren't fully free without wasting the whole channel).

■ **What does this mean in real life?** In ideal conditions, Wi-Fi 7 can reach staggering speeds – up to 46 Gbps theoretical on paper with 16 streams (no client laptop will do that; that's more for access points). But a more tangible number: a single-stream Wi-Fi 7 connection can be around 5.8 Gbps (320 MHz + 4K-QAM) versus 1.2 Gbps in Wi-Fi 6. That's roughly 4.8× faster. Now, in practice you likely won't get that unless you're very close to the router and using the 6 GHz band with no interference. However, even in crowded homes/offices, Wi-Fi 7 can significantly boost speeds or at least maintain high speeds more consistently. If you often transfer files between devices on Wi-Fi or have a multi-gigabit internet connection, Wi-Fi 7 can actually deliver those ultra-fast speeds wirelessly (Wi-Fi 6 might cap out around 0.8–1 Gbps real-world in many cases). Also, the latency improvements (up to 4x lower in best case) mean online gaming or video conferencing could feel more responsive and suffer fewer hiccups when the network is loaded.

■ **Crowded homes/offices:** Wi-Fi 7, with MLO, can excel in busy environments. For instance, in an apartment with tons of Wi-Fi networks, the 5 GHz band might be jammed. A Wi-Fi 7 laptop can use the 6 GHz band plus the 5 GHz simultaneously – 6 GHz might be clearer (because it's new and less used), and multi-link ensures even if one band gets interference, the other keeps things running smoothly. Think of it like having two internet lines combined – much less chance of a dropout. In an office, Wi-Fi 7's ability to service more devices (thanks to greater capacity and multi-user enhancements) means dozens of laptops can stream 4K or be in calls at once with fewer problems. It's more scalable – up to 5x the network capacity of Wi-Fi 6 in theory.

■ **Router readiness checklist:** Getting Wi-Fi 7 speeds isn't just about the laptop; your router/AP must support Wi-Fi 7, and your environment needs to allow it.

1. **6 GHz spectrum availability:** Wi-Fi 7 really shines with 6 GHz. Many countries opened 6 GHz for unlicensed use in Wi-Fi 6E. In India, historically only 2.4/5 were available, but as of mid-2025, India's government has moved to de-license the 5925–6425 MHz (lower 6 GHz) band for Wi-Fi. That means Wi-Fi 6E/7 devices can use that band indoors in India, which is great news (previously

## SHOULD YOU INSIST ON WI-FI 7 IN YOUR NEW LAPTOP?

In late 2024 and 2025, high-end models are starting to include Wi-Fi 7 modules (Intel's new Killer/Wi-Fi 7 540 modules, or Qualcomm Fast-Connect in ARM laptops). For example, the Dell XPS 13 Snapdragon X Elite in India includes Wi-Fi 7 capability. If you're buying a premium laptop that you plan to keep for, say, 4–5 years, getting Wi-Fi 7 is a nice future-proofing move – especially now that the spectrum is opening in India and elsewhere. It will fall back to Wi-Fi 6/5 with older routers, but when you do upgrade your network, you'll be ready. That said, if a laptop otherwise perfectly suits you but "only" has Wi-Fi 6 or 6E, it's not a deal-breaker for most in 2025. Wi-Fi 6 at 160 MHz can already hit ~1–1.5 Gbps in ideal conditions, which is plenty for common use. Wi-Fi 7 is more about pushing boundaries and ensuring low latency for future applications.

that band was off-limits). Check if your router supports 6 GHz and confirm regulations (they were in draft but likely to be finalized, making India the largest new 6 GHz market). If you're elsewhere, make sure 6E/7 is legal in your region – some places allow full 1200 MHz, some only 500 MHz (which means max 320 MHz channel might not fit alongside others).

2. **Wi-Fi 7 router or mesh system:** You'll need a Wi-Fi 7 capable router. Brands like TP-Link (Archer BE series), Asus (ROG/ZenWiFi with "BE" in name), Netgear (Orbi 970 series), etc., launched Wi-Fi 7 models. These are expensive as of 2025, but if you want to leverage your laptop's Wi-Fi 7, you should upgrade the router. Ensure the router has multigig uplink ports (2.5G or 10G Ethernet) if you have internet over 1 Gbps or plan heavy LAN transfers – otherwise the wireless speed is wasted bottlenecked by a 1 Gbps port.
3. **Channel configuration:** In a router's settings, you might need to enable 320 MHz channel width on 6 GHz for max speed. Also ensure it's running the latest firmware (early firmware might be buggy).
4. **Compatibility mode:** Wi-Fi 7 routers are backward compatible with Wi-Fi 6/5 etc. But sometimes combining old and new on the same band can slow things (e.g., legacy devices on 2.4 GHz). Use your router's ability to separate bands or smart connect features optimally. You might keep heavy usage devices on 6 GHz, and let IoT and older stuff use 5/2.4 GHz so they don't interfere.